PUC-SP

Helipad Detector

Rooftop Helipad Detection in Satellite Imagery of São Paulo with YOLO

Complete and Accessible Project Report

Technical pipeline, methodology, results, and a plain-language guide

Pontifical Catholic University of São Paulo (PUC-SP) — FACEI

BSc in Human Centered-AI & Data Science

Machine Learning / Computer Vision — Project P2

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Summary

This report works on two levels at once. For readers already familiar with data and AI, it is a complete technical report: it covers the 12-step pipeline, the three training experiments, the evaluation metrics, and the field validation across ten São Paulo neighborhoods. For readers who have never heard of YOLO or mAP, every section also includes a plain-language explanation — written as if we were simply talking through the project, with no prior AI knowledge assumed

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1. What Helipad Detector is, in one sentence

In one sentence: it's an Artificial Intelligence system that looks at satellite images of São Paulo and, on its own, points out where helipads exist on building rooftops.

In slightly more technical terms: the Helipad Detector is a complete Object Detection pipeline that locates helipads on rooftops in the city of São Paulo, using high-resolution aerial/orbital imagery and models from the YOLOv8/YOLOv11 family. The project covers the full lifecycle of an applied computer vision system: programmatic image collection, geospatial automation, building and curating an original dataset, annotation, preprocessing, training, quantitative and qualitative evaluation, inference on neighborhoods unseen during training, an extra field-validation stage on rea

> An important point for non-technical readers: the project did not use any pre-made image bank. The entire data foundation — the satellite photos and the markings of where the helipads are — was built from scratch by the team, a stage that usually consumes most of the effort in any serious Artificial Intelligence project.

2. Why detect helipads — and why it's hard

Helipads were chosen as the target because, seen from above, they have a very distinctive design: a large “H” painted over a circular or square area on the rooftop. That sounds easy to recognize — but in practice it isn't, and that very difficulty is what makes the project interesting from an educational standpoint.

In satellite imagery of a dense city like São Paulo, the model can get confused with:

- Building swimming pools, which have a shape and contrast similar to the “H”;
- Sports courts, with lines and geometric markings that look alike;
- Rooftop structures, water tanks, and air-conditioning equipment;
- Reflective surfaces, which change appearance depending on daylight.

This kind of confusion is called, in the field, a false positive — when the model “sees” a helipad where none exists. Working with a target that generates plausible false positives is what makes this project a good computer vision exercise: it forces the team to think about annotation quality, example diversity, and the model's real limits, instead of celebrating a nice-looking number without understanding what's behind it.

3. How an AI learns to "see" — a plain-language primer

What is "Object Detection"?

What is YOLO?

How does the model "learn"?

Why does data quality matter more than the model itself?

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> *This kind of error is known as a data-integrity issue: the code ran without crashing, the numbers looked plausible, but the experiment attribution was wrong. In any real AI project, this is exactly the most dangerous type of silent failure — because it doesn't raise a visible error, only a false conclusion.*

8. Quantitative evaluation — what the numbers mean

8.1 Results of the three experiments

Experiment	Best Epoch	Precision	Recall	mAP@50	mAP@50-95
exp1 (60 ep., original dataset)	54	1.000	0.963	0.994	0.881
exp2 (100 ep., original dataset)	81	0.982	0.971	0.993	0.888
exp3 (100 ep., augmented fork)	78	0.943	1.000	0.993	0.885

8.2 Training charts — exp1 (60 epochs, original dataset)

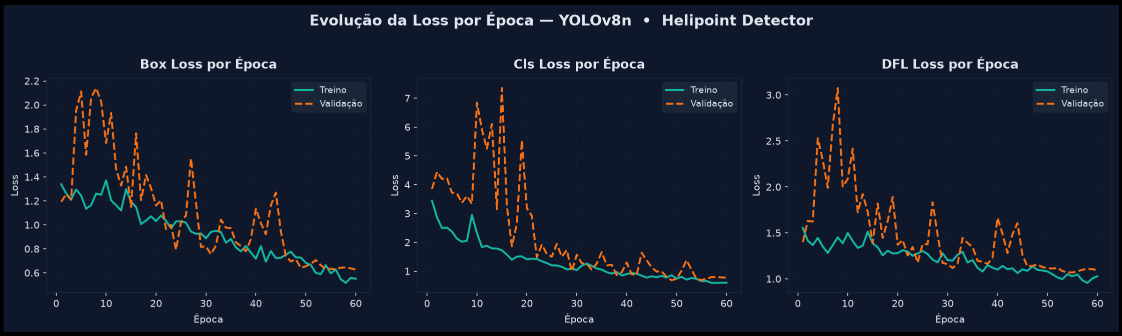


Figure 1 — Evolution of the losses (Box, Cls, DFL) per epoch, train vs. validation. No clear sign of overfitting across the 60 epochs; val/cls_loss oscillates in the first 20 epochs and stabilizes from epoch 30 onward.

Figure 2 — Precision and Recall over training. Precision stabilizes above 0.95 from epoch 30 onward; Recall reaches 0.97+ in the final epochs, after a sharp dip around epoch 5.

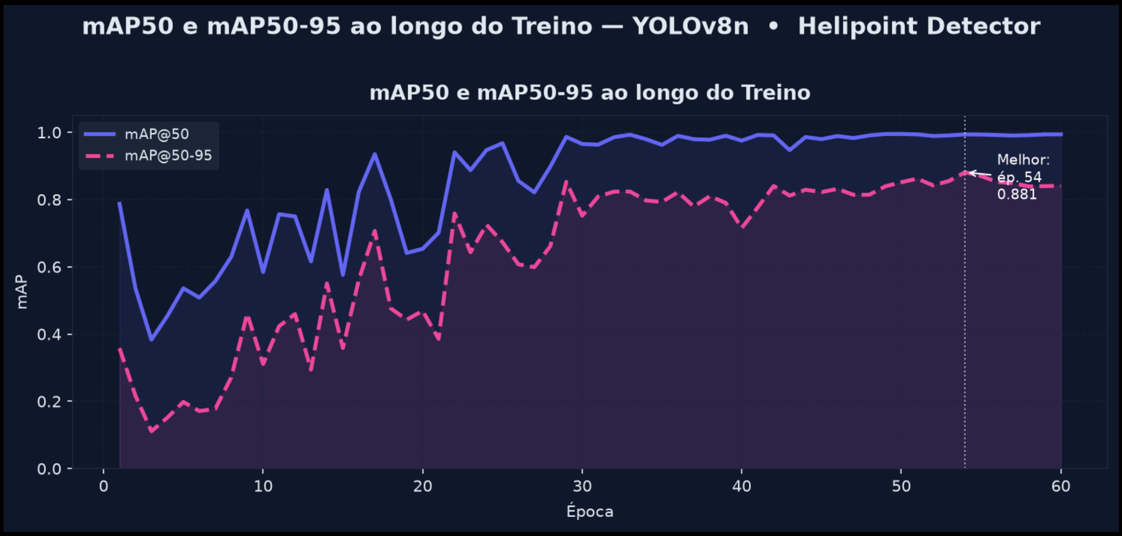
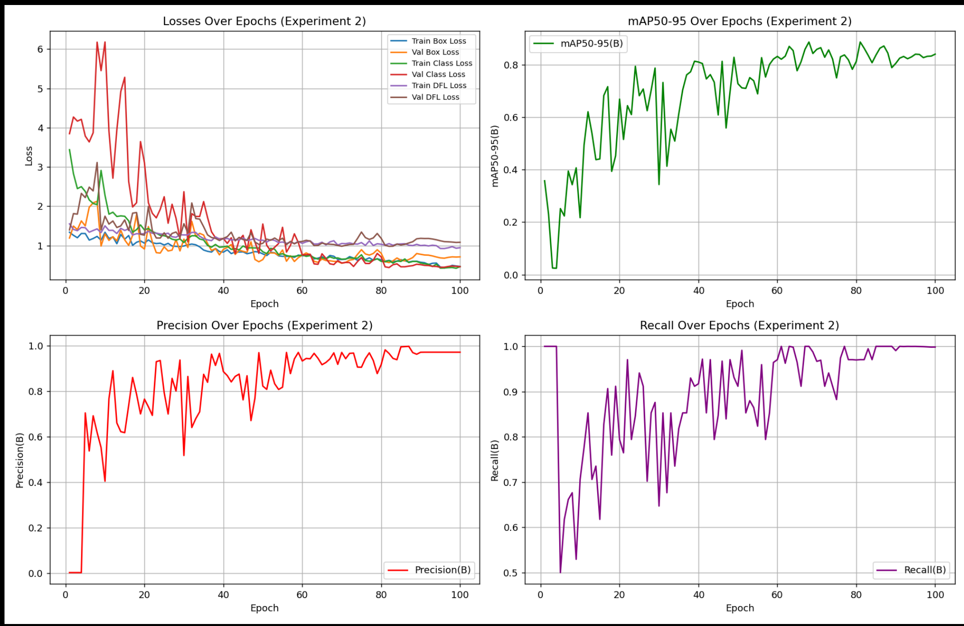
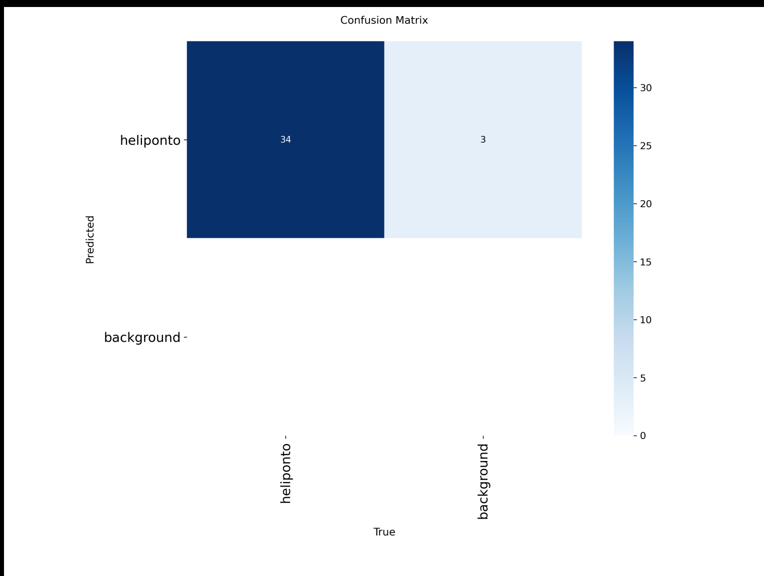


Figure 3 — mAP@50 and mAP@50-95. Peak at the best epoch (54), after which the metrics fluctuate slightly.

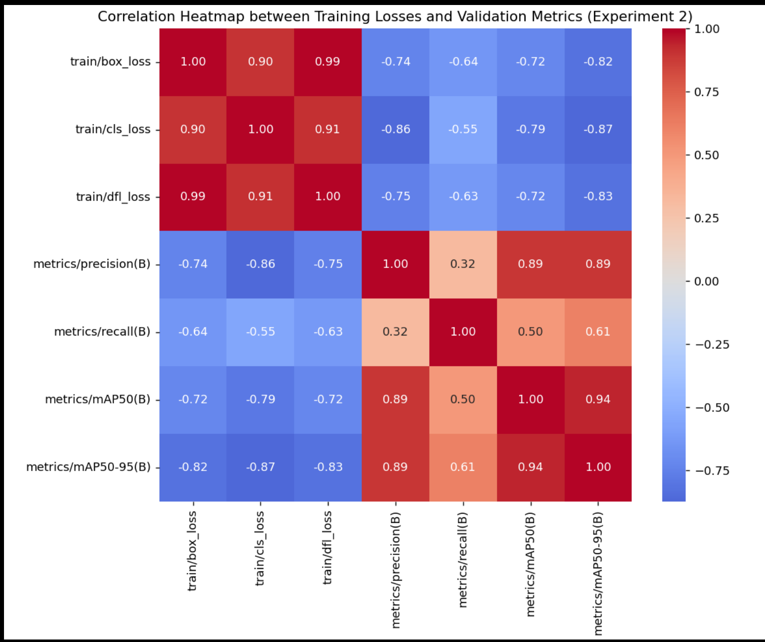
8.3 Training charts — exp2 (100 epochs, original dataset)



Overview — losses, mAP@50-95, Precision, and Recall across 100 epochs (exp2).

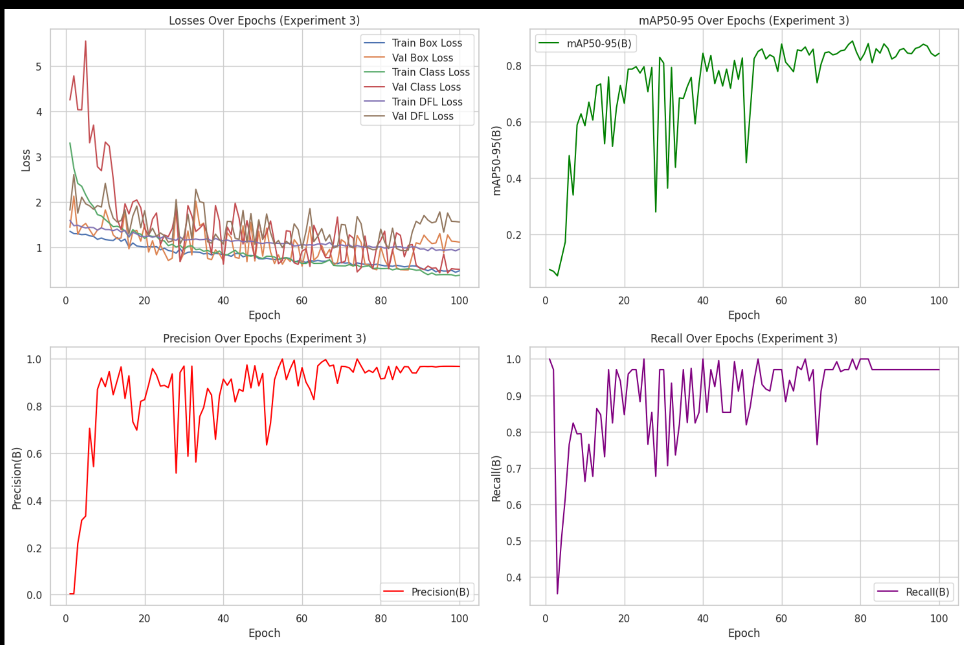


Confusion matrix (validation, exp2): 34 hits, 3 background false positives confused with a helipad.

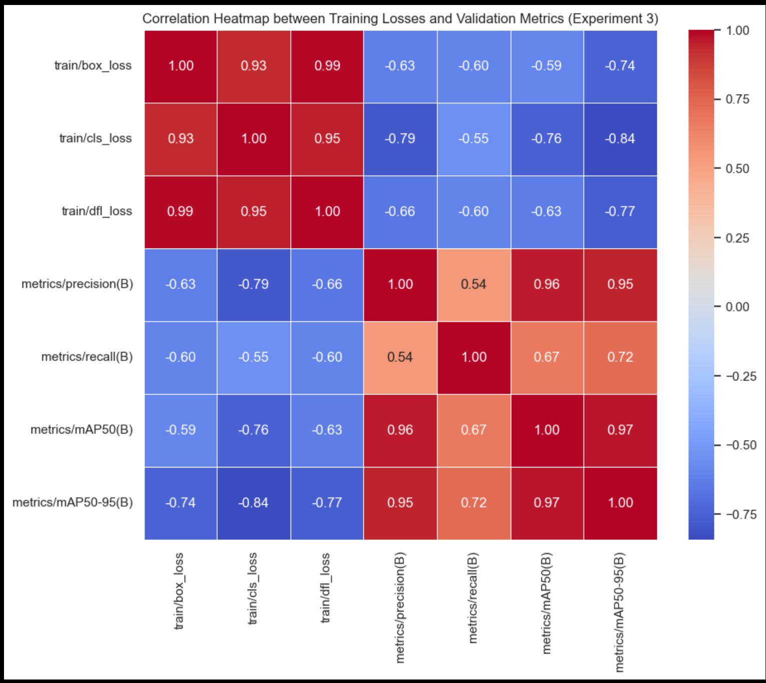


Correlation between training losses and validation metrics (exp2).

8.4 Training charts — exp3 (100 epochs, forked/augmented dataset)



Overview — losses, mAP@50-95, Precision, and Recall across 100 epochs (exp3). Same pattern of a slight dip between the best epoch and the final one observed in exp1.



Correlation between training losses and validation metrics (exp3): cls_loss is the most strongly (anti-)correlated with mAP@50-95 (-0.84).

9. Field validation — testing on real data across ten neighborhoods

Neighborhood	Detected	Total Tiles	Rate
Inter-Zone Corridor	133	480	27.7%
Itaim Bibi	191	750	25.5%
Brooklin	231	1,014	22.8%
Vila Olímpia	158	660	23.9%
Paulista Avenue (Section 1)	179	840	21.3%
Faria Lima	170	840	20.2%
Paulista Avenue (Section 2)	157	780	20.1%
Pinheiros	244	1,232	19.8%
Vila Nova Conceição	109	572	19.1%
Alphaville Industrial	105	775	13.6%
TOTAL	**1,677**	**7,943**	**21.1%**

10. Qualitative analysis — hits, misses, and why

Type	Example Tile	Confidence	Note
Clear hit	tile_z19_x194126_y297485.jpg	0.94	Sharp “H” pattern in a well-defined square on the rooftop
Clear hit	tile_z19_x194143_y297481.jpg	0.96	Clear geometry and contrast, well-fitted box
Challenging hit	tile_z19_x194545_y298183.jpg	0.77	Correct detection despite a less favorable angle/lighting
False Positive	tile_z19_x194129_y297480.jpg	0.78	Box over a sports court, not a helipad
False Positive	tile_z19_x194547_y298176.jpg	0.86	Box over a swimming pool, geometrically similar to the “H”
Data failure	tile_z19_x194139_y297467.jpg	—	Empty/black tile (ESRI download failure) flagged as “Detected”

10.1 Real detection examples (curated test set, exp1)

Clear hits (high confidence)

helipad — confidence 0.95. Sharp “H” pattern, box well fitted to the helipad's outline.

helipad — confidence 0.86. Clear marking identified even with vegetation and shadow near the rooftop.

Challenging hit

helipad — confidence 0.64. Moderate confidence; the helipad structure is visually less evident in this capture.

False positives

helipad — confidence 0.28. Ground/road marking confused with a helipad pattern — low confidence, consistent with a false positive.

helipad — confidence 0.28. Another case of a road/pavement generating a false positive, again with low confidence.

11. Interactive web application

Tab	Purpose
Experiment Metrics	Automatically discovers each trained experiment, comparing Precision/Recall/mAP live
Field Detections by Region	Cards, bar chart, and table of the field validation across 10 neighborhoods
Map	Dark-mode Folium map with 3 switchable layers, including a blue-scale detection-rate layer
Search by Region	Live bounding-box search: downloads tiles and runs inference on demand
Sample Images	Pre-selected real detections for instant demonstration
Upload Image	Upload any aerial/satellite image for annotated detection
Pipeline	Visual step-by-step of the complete 12-step pipeline
Governance	Fairness, LGPD, and known-limitations statements
Downloads	Reports, dataset, metrics, and field-validation artifacts

12. Strengths, limitations, and next steps

12.1 Strengths

12.2 Observed limitations

12.3 Suggested future improvements

13. Ethics, LGPD, and governance

13.1 A security incident identified and fixed

> *In plain terms: a “password” granting access to Roboflow was visible to anyone who opened the notebook — including in a public GitHub repository. That's a real risk, because anyone who saw that key could use it to access the project's workspace. The team fixed the problem by moving the key out of the code — a concrete example of why good secrets-management practices matter even in academic projects, not only in commercial products.*

14. Conclusion

15. Glossary of technical terms

Term	Plain-language meaning
Bounding box	A rectangular box drawn around an object in an image, indicating where it is.
Dataset	The set of data (here, images + annotations) used to train and test the model.
Epoch	One complete pass of the model through all the training images.
False Negative (FN)	When a real helipad exists in the image, but the model fails to detect it.
False Positive (FP)	When the model detects a helipad where none actually exists.
Generalization	The model's ability to perform well on new images it has never seen during training.
IoU (Intersection over Union)	A measure of how much the model's predicted box overlaps with the object's real box.
mAP (mean Average Precision)	A single score summarizing overall detection quality across several overlap thresholds.
Overfitting	When the model “memorizes” the training examples instead of learning the general pattern, performing poorly on new data.
Precision	Of all the detections made, how many were correct.
Recall	Of all the real objects that exist, how many the model managed to find.
Tile	A standardized square piece of a map/satellite image, used to assemble larger mosaics.
YOLO	“You Only Look Once” — a family of AI models for real-time object detection.

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Report compiled from the project's full README, the results analysis (Analysis_yolo_results.md), the technical PDF documentation, and the team's executive summary (PT and EN).